Cost Variance (CV) = EV - AC

Schedule Variance (SV) = EV - PV

cost performance index (CPI) = EV ÷ AC

Schedule Performance Index (SPI) = EV ÷ PV

Estimate at Completion (EAC) = BAC ÷ CPI

Estimate to Complete (ETC) = EAC - AC

Variance at Completion (VAC) = BAC - EAC

Set: 3. Marketing & Branding Campaigns

Case A - Global product launch digital media buy

Given:

PV = $95,000

EV = $95,000

AC = $115,000

BAC = $250,000

Metric	Calculation	Answer
CV	95000 - 115000	-20,000
SV	95000 - 95000	0
CPI	95000 ÷ 115000	0.83
SPI	95000 ÷ 95000	1
EAC	250000 ÷ 0.8261	302,632
ETC	302,632 - 115,000	187,632
VAC	250,000 - 302,632	-52632.

The project is on schedule (SPI = 1.00) but over budget (CPI < 1).

Case -B - Influencer outreach & content product

Given

PV = 30,000
EV = 35,000
AC = 31,000
BAC = 90,000

Metric	Calculation	Answer
CV	35000 - 31000	4000
SV	35000 - 30000	5000
CPI	35000 ÷ 31000	1.13
SPI	35000 ÷ 30000	1.17
FAC	90000 ÷ 1.1290	79714
ETC	79714 - 31000	48714
VAC	90000 - 79714	10,286

The project is ahead of schedule (SPI > 1) and under budget (CPI > 1).

Case-C - TV comercial post-production

Given

PV = 150,000
EV = 120,000
AC = 140000
BAC = 400,000

Metric	Calculation	Answer
CV	120,000 - 140,000	- 20,000
SV	120,000 - 150,000	- 30,000
CPI	120,000 ÷ 140,000	0.86
SPI	120,000 ÷ 150,000	0.80
EAC	400,000 ÷ 0.857	466,667
ETC	466,667 - 140,000	326,667
VAC	400,000 - 466,667	-66,667

The project is behind schedule (SPI < 1) and over budget (CPI < 1).

Comparison

Case	CV	SV	CPI	SPI	Status
A	-$20,000	$0	0.83	1.00	over budget, on Schedule
B	+$40 000	+$5,000	1.13	1.17	under budget, ahead of schedule
C	-$20000	-30000	0.86	0.80	over budget, behind Schedule

Rules

CV > 0 → under budget
CV < 0 → over budget
SV > 0 → ahead of schedule
SV < 0 → behind schedule
CPI / SPI > 1 → favorable
CPI / SPI < 1 → unfavorable

Purpose of CPI, SPI, EAC

PV, EV, AC → Current Situation

PV - Planned values measures how much work should have been completed by now according to the plan.

EV - Earned Value measures the budgeted Value of the work actually completed.

AC - Actual cost measures how much money has actually been spent on the completed work.

CV, SV → Variance

CV - Cost Variance compares earned value with ~~planned~~ actual cost to determine whether the project is over or under budget.

SV - Schedule Variance compares earned value with planned value to determine whether the project is ahead or behind schedule.

CPI, SPI → performance

CPI → Cost performance index measures cost efficiency. How efficiently are we using money,

SPI → Schedule performance index measures schedule efficiency. How efficiently are we progressing?

EAC, ETC, VAC → Future

EAC - Estimate at completion predicts the expected total cost when the project finishes. How much will the project finally cost?

ETC → Estimate to complete estimates the additional money required to finish the project. How much more money do we need?

VAC → Variance and completion compares the original budget with the predicted final cost. "Will we finish over or under the original budget?"